Customer Churn Prediction and Dynamic Pricing Optimization

Milestone: Data Collection, Data Visualization, Data Exploration and Data Processing

Group 16

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Introduction

Customer churn prediction has become a critical analytical task for modern subscription-based businesses. Companies that operate on recurring revenue models, such as streaming platforms, rely heavily on customer retention to sustain profitability and long-term growth. Even a small increase in churn rate can significantly reduce revenue, making it essential for organizations to understand why customers leave and how to prevent it.

This project focuses specifically on Netflix customer churn prediction because Netflix represents one of the most prominent global subscription platforms and provides a realistic case study for analyzing user behavior patterns. Streaming services like Netflix generate extensive behavioral data, including watch time, engagement levels, satisfaction scores, and subscription duration. Such rich datasets make it possible to apply data mining and machine learning techniques to uncover patterns associated with churn.

Netflix was chosen for this analysis for several reasons:

- It operates on a subscription-based model where retention is directly tied to revenue.
- User engagement data is highly measurable, and behavior driven.
- Customer decisions to stay or leave are influenced by multiple factors such as pricing, content satisfaction, and usage patterns.

It provides a practical scenario for applying predictive analytics techniques commonly used in industry.

By analyzing Netflix customer data, this study aims to identify behavioral and demographic characteristics that influence churn, understand relationships between engagement and retention, and build a foundation for predictive modeling. Insights obtained from this analysis can help streaming platforms design targeted retention strategies, optimize pricing, improve customer satisfaction, and ultimately enhance user loyalty.

Overall, this project will demonstrate how exploratory data analysis, visualization, statistical testing, and preprocessing techniques can be used to transform raw customer data into actionable business intelligence.

Data Collection

1.1 Dataset Description

This project uses a structured Netflix customer churn dataset designed to support customer-retention analytics in a subscription-based streaming environment. The dataset includes demographic details, subscription type, engagement behavior, payment preferences, and content taste indicators. It was imported and analyzed using Python-based tools such as Pandas and NumPy for data handling, Matplotlib/Seaborn for visualization, and Scikit-learn for preprocessing and baseline modeling.
Each row represents one customer and captures a snapshot of their interaction with Netflix (e.g., watch hours, average daily watch time, and recency of login). The objective is to uncover measurable drivers of churn and build an initial predictive model that can support retention strategies.

1.2 Dataset Characteristics

- Total Records: 5,000 customers
- Total Features: 14 variables
- Target Variable: churned (1 = churned, 0 = retained)
- Data Types: Mixed numeric and categorical features
- Practical Value: Suitable for churn classification, customer profiling, and retention strategy design

1.3 Key Attributes Collected

- Demographics & Identity: Age, Gender, Region
- Subscription & Pricing: Subscription Type, Monthly Fee, Number of Profiles
- Behavioral metrics: Watch Hours, Average Watch Time Per Day, Last Login Day
- Content & device preferences: Favorite Genre, Device
- Financial behavior: Payment Method, Number of Profiles
- Target variable: Churned (1 = Yes, 0 = No)

1.4 Suitability for Analysis

This dataset supports churn analysis because it contains strong churn-linked categories: (1) engagement intensity, (2) inactivity/recency, and (3) subscription and pricing context. These variables reflect common real-world churn drivers such as declining usage, reduced perceived value, and preference mismatches.

Exploratory Data Analysis (EDA)

EDA was conducted to understand the dataset's structure, identify anomalies, examine feature distributions, and observe relationships that may influence churn. Each visualization below was generated from the actual data and is described in detail.

2.1 Missing Values Analysis

The missing-value heatmap confirms that the dataset contains no null entries across all columns. Every customer record includes complete demographic, behavioral, and subscription information.
This represents a highly favorable data quality condition because no imputation strategies were required and no observations needed removal. Complete data improves statistical reliability and ensures the model learns from the full feature space without artificial assumptions introduced through missing-value replacement.

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No missing values found in the DataFrame.
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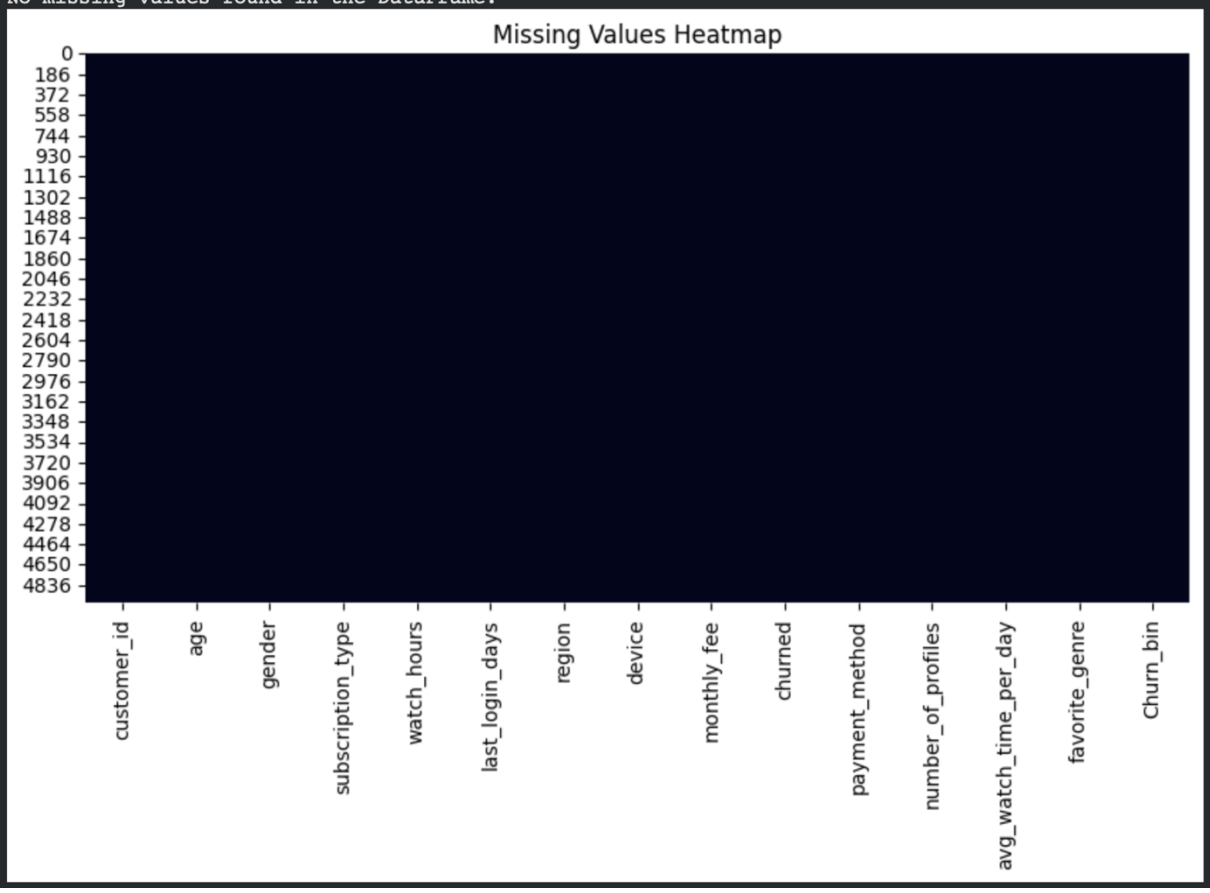
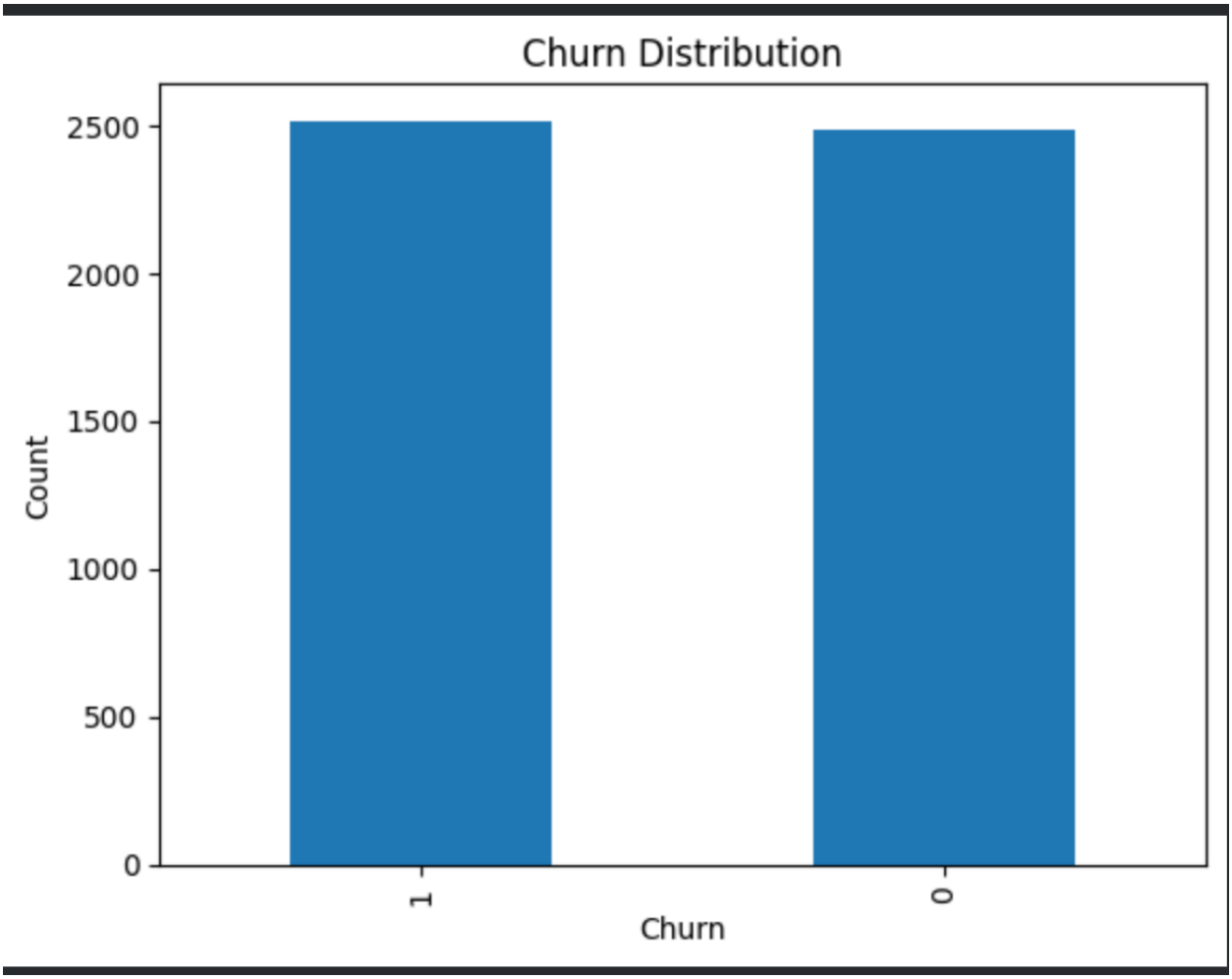


2.2 Target Variable Distribution (Class Balance)

The churn distribution bar chart shows nearly equal representation between churned and retained customers, with approximately 502 churned users and 498 retained users. Balanced datasets are ideal for classification modeling because algorithms learn patterns equally from both outcomes. Accuracy becomes a meaningful performance metric, and additional balancing techniques such as oversampling or SMOTE are unnecessary at this stage.



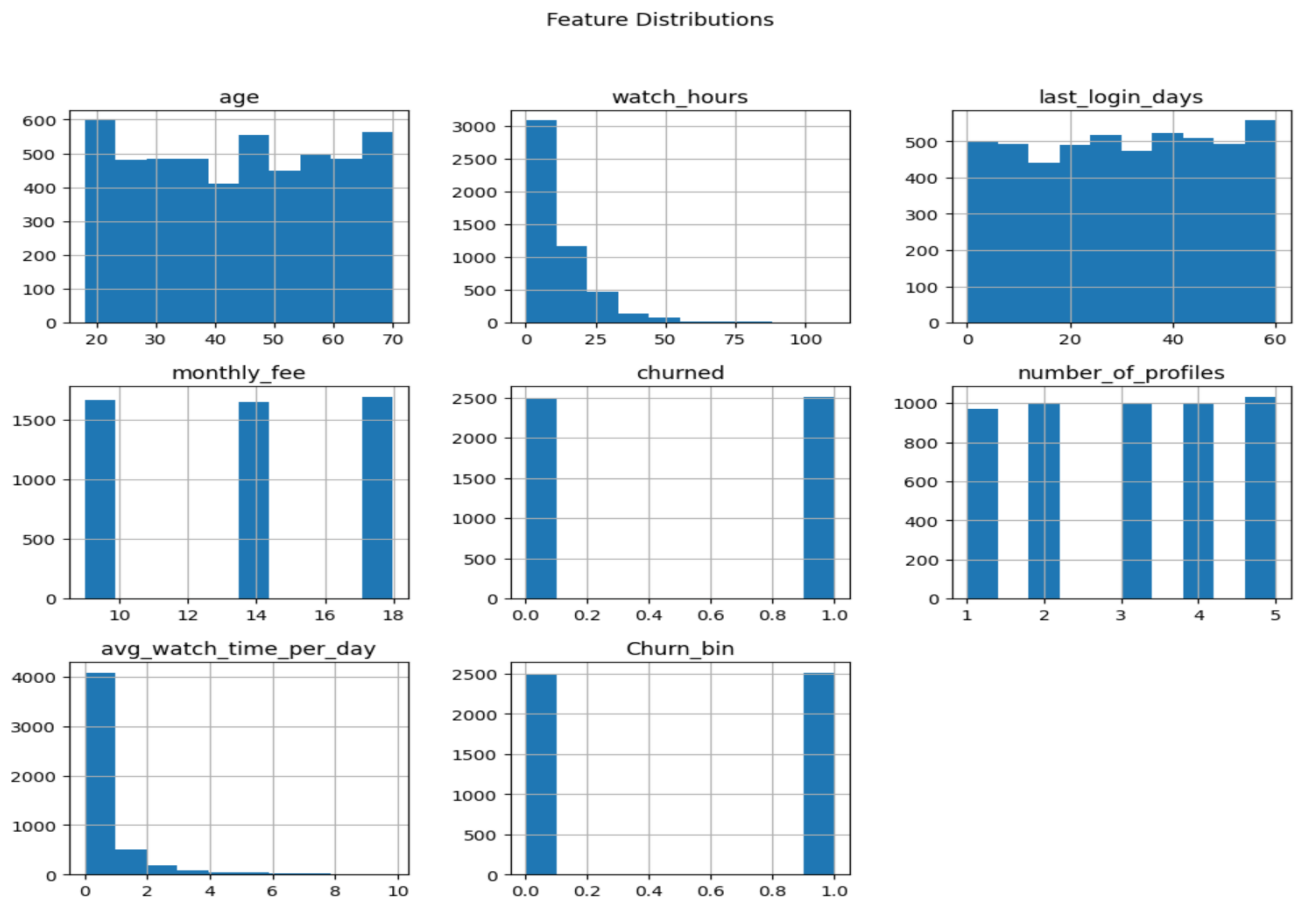
Analytical Insight:
Balanced class representation reduces prediction bias and strengthens model evaluation validity.

2.3 Distribution of Numerical Variables
Histogram analysis reveals several important behavioral and subscription patterns:

- Age distribution spans the full adult range, suggesting broad demographic coverage.

- Watch Hours and Average Watch Time Per Day exhibit right-skewed distributions, indicating most moderate viewers alongside a smaller group of heavy users.
- Last Login Days varies widely from recently active users to customers inactive for extended periods, providing a strong behavioral signal for churn risk.
- Monthly Fee displays discrete peaks aligned with Netflix subscription tiers.

Behavioral engagement variables demonstrate significantly greater variability compared to demographic attributes.

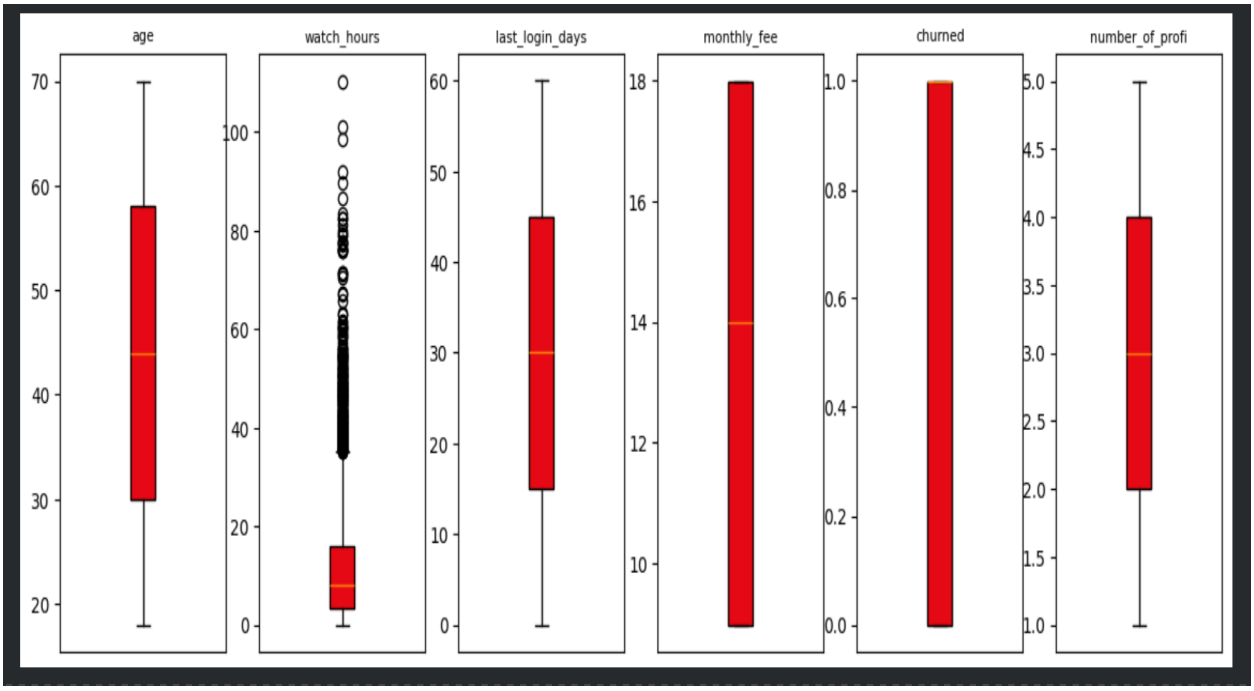


Analytical Insight:
Customer engagement patterns provide richer predictive signal compared to static demographic factors.

2.4 Outlier Detection
Boxplots were used to identify extreme values within each numerical feature.

Key findings:

- Watch Hours and Average Watch Time Per Day showed the most notable outliers — a small subset of users with extremely high values (e.g., 80–98 hours). These represent genuinely heavy users rather than data entry errors, but they could act as leverage points that distort model coefficients if not treated.
- Last Login Days had a clean distribution with few outliers, ranging naturally from 0 to 60.
- Age and Monthly Fee showed no significant outliers, remaining within expected realistic ranges.
- Number of Profiles also remained cleanly within the 1–5 range with no anomalies.

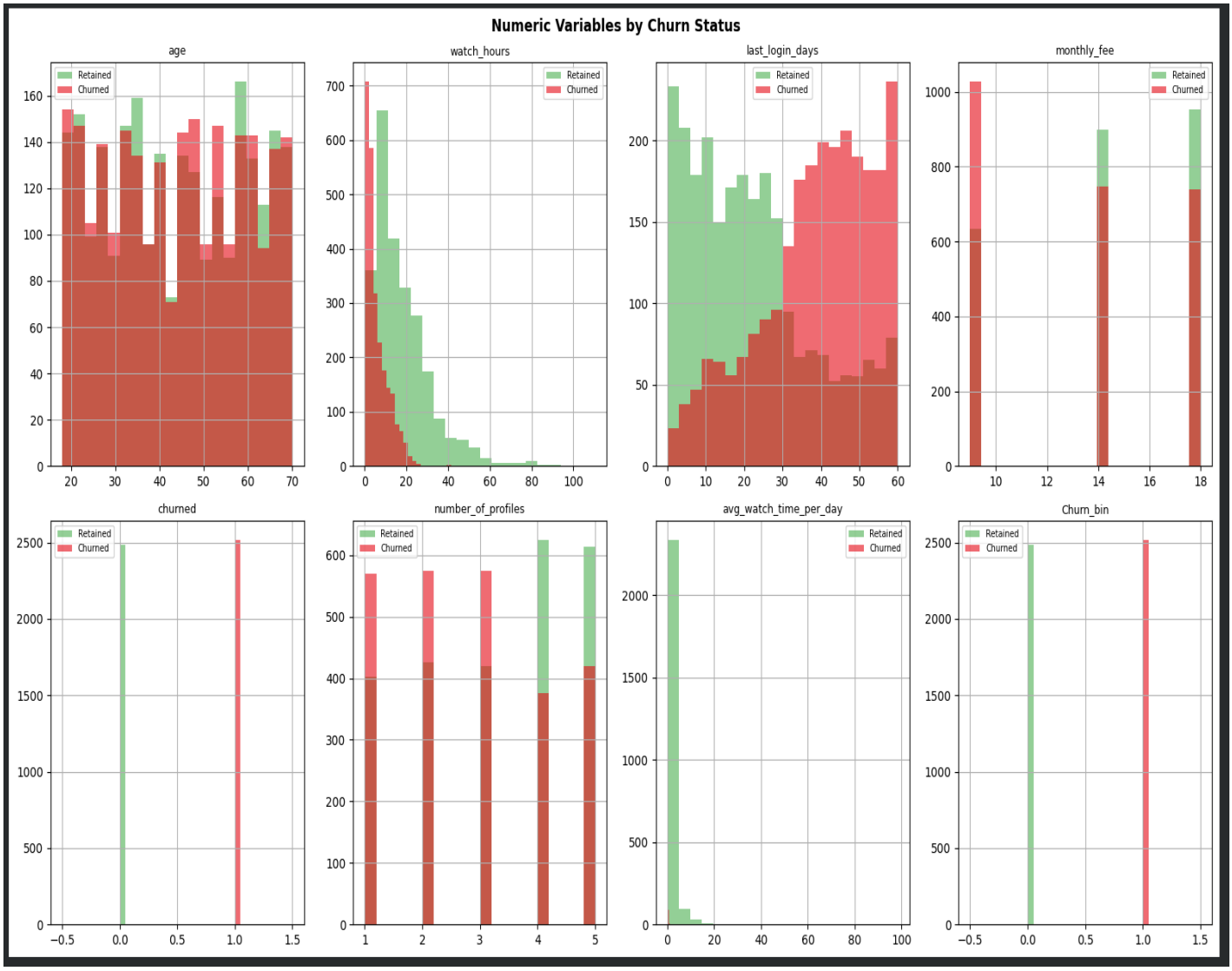


Analytical Insight:
Outliers in watch behavior are flagged for potential capping or log-transformation in advanced modeling stages but were retained for baseline modeling.

2.5 Numerical Variables Split by Churn Status
Overlayed histograms comparing churned and retained customers provide direct insight into churn drivers.

Key findings include:
- Last Login Days shows clear separation — churned customers demonstrate longer inactivity periods prior to cancellation.

- Higher Watch Hours and Average Watch Time Per Day are more concentrated among retained users.
- Age distributions overlap heavily, indicating weak predictive influence.
- Monthly Fee shows minimal separation between groups.

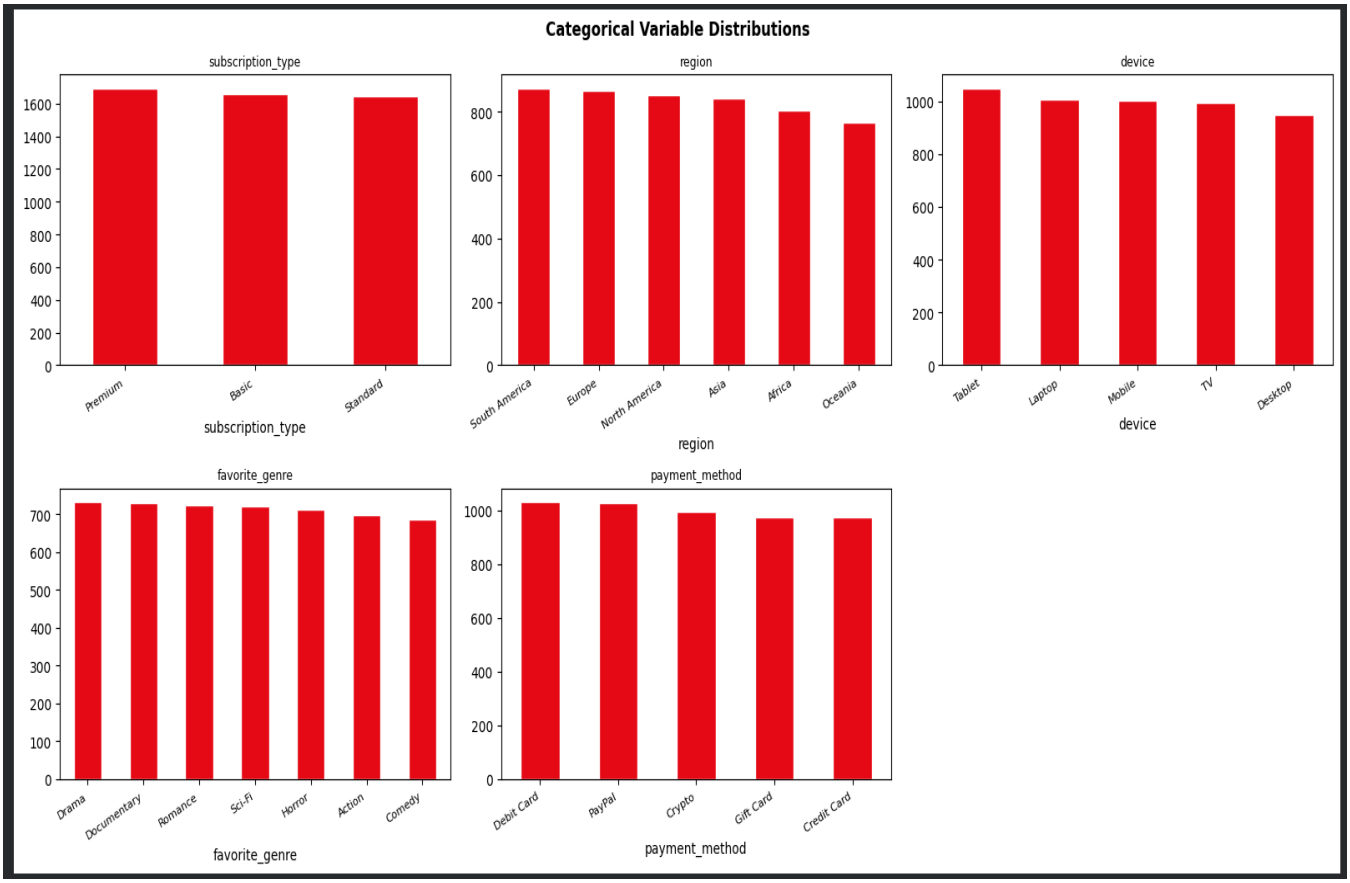


2.6 Categorical Variable Distributions

This grid of bar charts shows the frequency of each category within the five main categorical variables. Observations:

- Subscription Type is approximately evenly split across Basic, Standard, and Premium, indicating no single plan dominates the dataset.
- Region is also evenly distributed across Asia, Africa, Europe, North America, Oceania, and South America, confirming global coverage.

- Device shows relatively equal representation across TV, Mobile, Laptop, Tablet, and Desktop.
- Favorite Genre is spread across Action, Comedy, Documentary, Drama, Horror, Romance, and Sci-Fi with no single genre dominating.
- Payment Method shows a roughly even distribution across Crypto, Credit Card, Debit Card, Gift Card, and PayPal.



The balanced distribution across all categorical variables indicates that the dataset is well-sampled and unlikely to introduce representation bias from any single category.

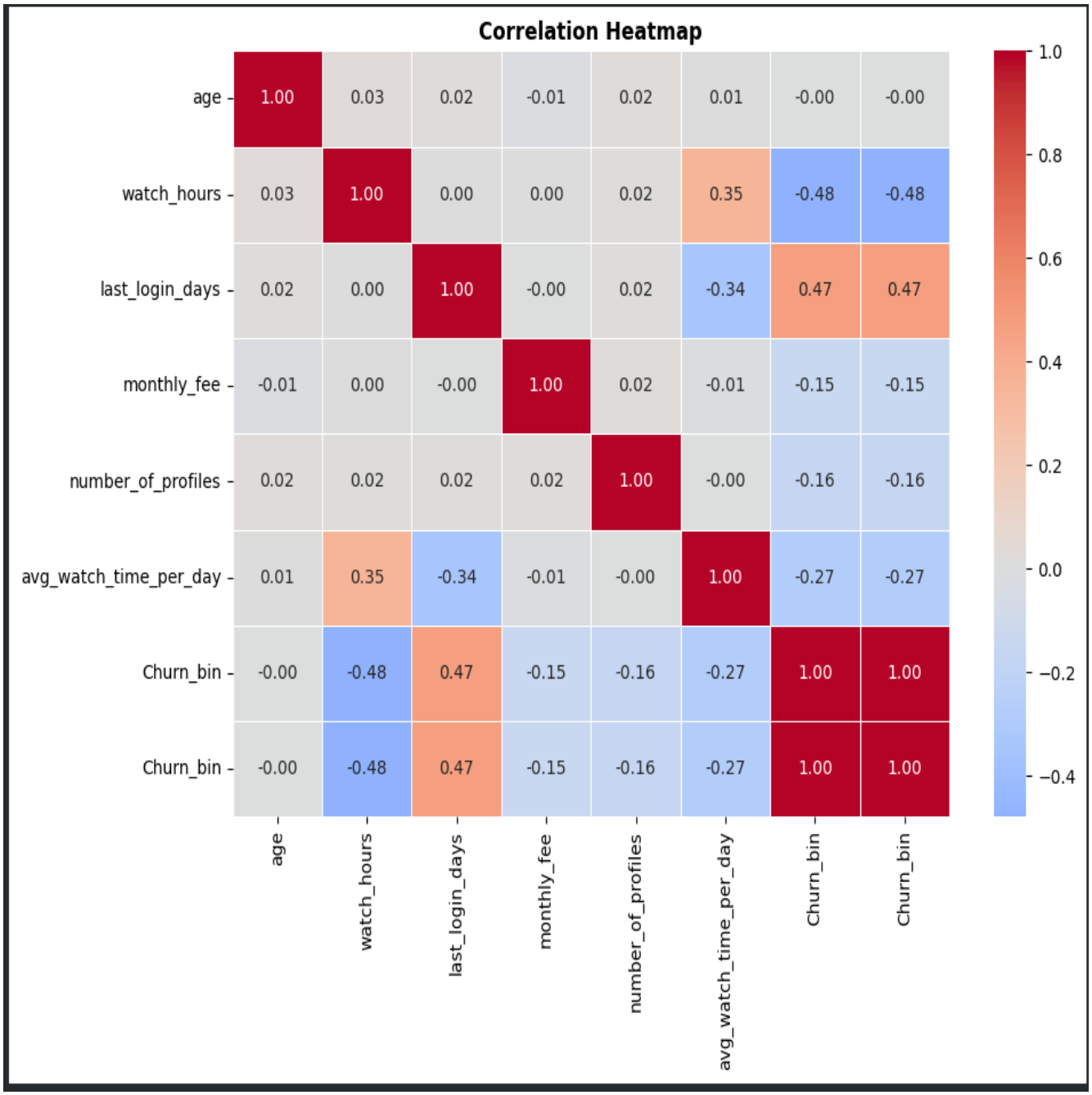
2.7 Correlation Heatmap

Correlation analysis demonstrates mostly weak-to-moderate relationships between variables.

Important observations include:

- Negative correlation between churn and engagement metrics confirms higher viewing activity reduces churn likelihood.
- Positive correlation between churn and last login days supports inactivity as a churn signal.

- Limited multicollinearity indicates features contribute distinct predictive information.



Data Processing and Preprocessing

3.1 Data Cleaning

Data cleaning procedures were performed to ensure dataset reliability and modeling readiness. Duplicate records were identified and removed to prevent repeated customer profiles from introducing bias during model training. Data types were validated and corrected so that numerical attributes remained in appropriate numeric formats while categorical variables were consistently represented. Range validation checks were also conducted to confirm that values such as age, login recency, and subscription-related features remained within realistic operational limits. Extreme values observed in the Average Watch Time per Day column, particularly those exceeding 10 hours, were treated as likely decimal-point entry errors and corrected by dividing affected values by ten. Category labels were standardized by removing extra whitespace and formatting inconsistencies to prevent duplicate category creation during encoding. Since the dataset contained no missing values, imputation techniques were not required, allowing all observations to be retained for analysis.

3.2 Encoding Categorical Variables

Categorical variables including gender, region, device type, subscription plan, payment method, and favorite genre were transformed into numerical format using one-hot encoding through pd.get_dummies() with drop_first=True. This encoding strategy converts each category into binary indicator variables while avoiding artificial ordinal relationships between categories. The removal of one reference category prevents the dummy variable trap and reduces multicollinearity in linear models such as Logistic Regression. Prior to encoding, categorical values were standardized to maintain consistency across labels. Although one-hot encoding increases dataset dimensionality, feature expansion was monitored to maintain computational efficiency and interpretability. The encoding process was applied consistently across training and testing datasets to prevent category mismatch and ensure reproducible model performance.

3.3 Feature Scaling

Feature scaling was applied to numerical variables using StandardScaler to standardize feature distributions before Logistic Regression modeling. Standardization centers each variable around a mean of zero with unit variance, ensuring that features measured on larger numerical ranges, such as watch hours or login days, do not dominate smaller-scale attributes during optimization. Scaling improves gradient convergence speed and stabilizes coefficient estimation, leading to more reliable decision boundaries. To prevent data leakage, scaling parameters were learned only from the training dataset and subsequently applied to the testing data. Tree-based algorithms evaluated in later

stages, including Decision Trees and Random Forest models, do not require scaling because their split-based learning mechanisms are insensitive to feature magnitude differences.

Baseline Modeling and Evaluation

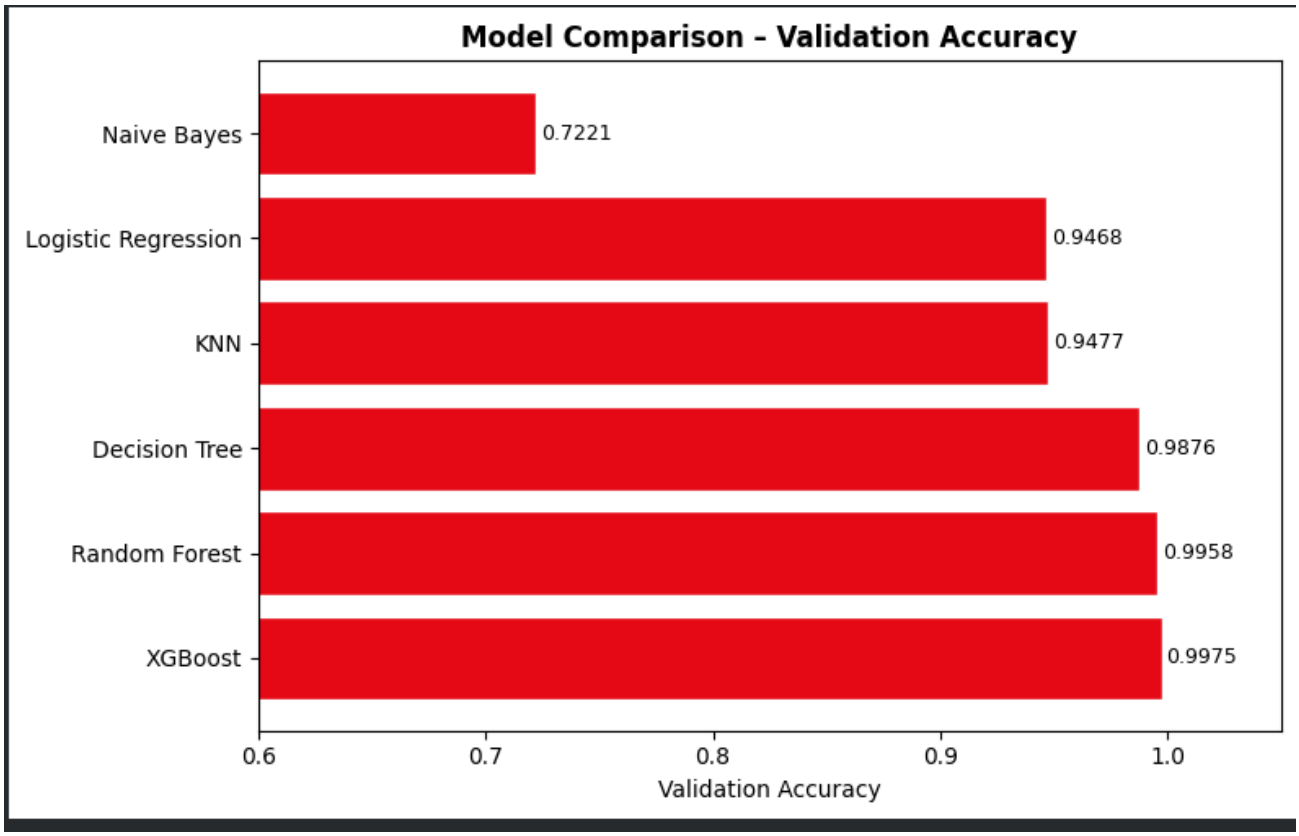
4.1 Model Selection

A Logistic Regression classifier was selected as the baseline predictive model due to its interpretability, computational efficiency, and suitability for binary classification problems such as churn prediction. Logistic Regression provides clear insight into feature influence through coefficient interpretation and serves as an effective benchmark for evaluating more complex machine learning algorithms. The dataset was divided into training and testing subsets using an 80/20 stratified split to preserve churn class balance across both sets. Following preprocessing, encoded categorical variables and standardized numerical features were used to train the model, ensuring consistency between training and evaluation phases.

4.2 Model Comparison

To evaluate alternative modeling approaches, multiple supervised classification algorithms were tested and compared using validation accuracy as the performance metric. The evaluated models included XGBoost, Random Forest, Decision Tree, K-Nearest Neighbors, Logistic Regression, and Naive Bayes.

Tree-based ensemble models demonstrated superior performance compared to linear and probabilistic approaches. XGBoost achieved the highest validation accuracy, followed closely by Random Forest and Decision Tree classifiers. These results indicate that customer churn behavior contains nonlinear relationships and complex feature interactions that ensemble learning methods can capture more effectively. Logistic Regression and KNN achieved competitive performance, confirming that linear and distance-based patterns also contribute meaningful predictive signal. Naive Bayes performed comparatively poorly due to violations of its feature independence assumption, particularly among correlated engagement variables such as watch time and login recency.

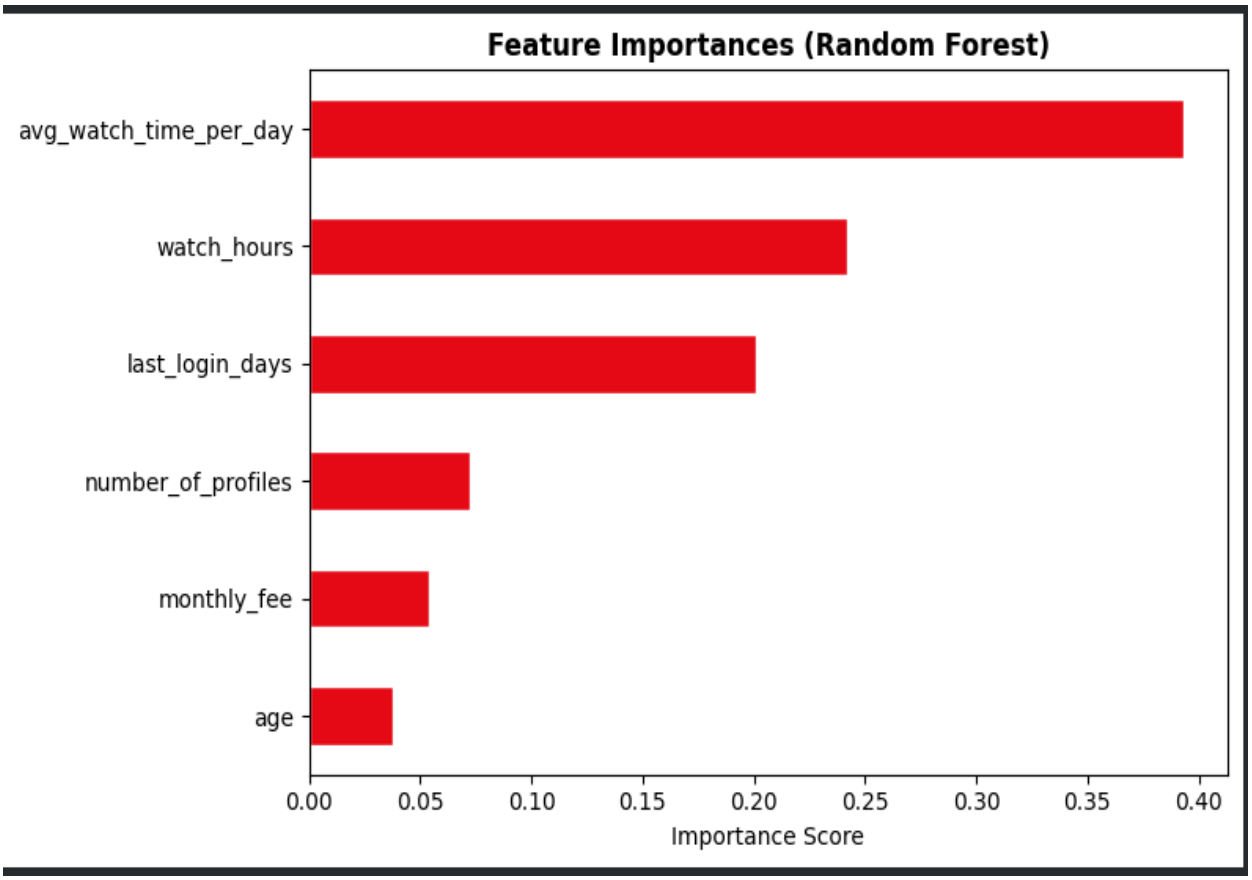


4.3 Feature Importance Analysis

Feature importance scores were extracted using the Random Forest classifier to identify the most influential predictors of customer churn. The analysis revealed that Last Login Days is one of the strongest drivers of churn probability, indicating that prolonged inactivity significantly increases cancellation risk. Engagement-related features such as Average Watch Time Per Day and Watch Hours also ranked highly, confirming that reduced platform usage is strongly associated with churn behavior.

Subscription-related variables, including Monthly Fee, contributed moderately to prediction performance, suggesting that pricing tier and perceived value may influence retention decisions. Demographic attributes such as Age and Number of Profiles showed smaller but non-zero contributions, indicating supporting influence rather than primary predictive power. Encoded categorical features such as payment method and device preference provided additional contextual signals but were comparatively less dominant.

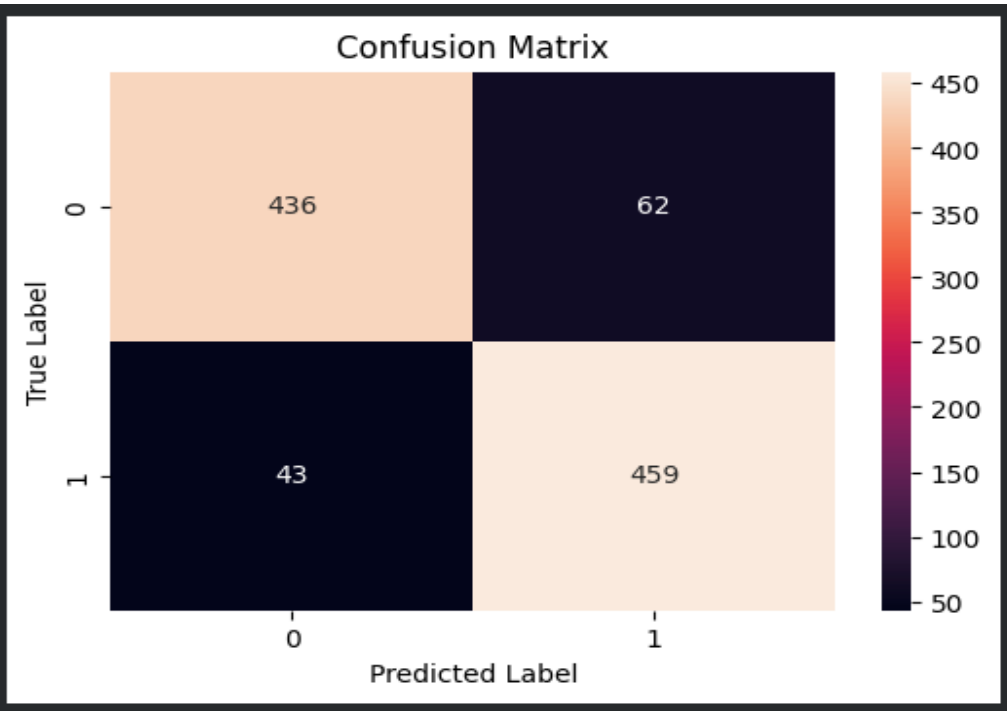
From a business perspective, these findings suggest that monitoring inactivity and declining engagement could serve as effective early-warning indicators for targeted retention interventions.



4.4 Confusion Matrix Evaluation

The confusion matrix summarizes the Logistic Regression model's predictive performance on the held-out test dataset. The model correctly classified 436 retained customers as non-churners (true negatives) and 459 churned customers as churners (true positives). Misclassification errors consisted of 62 false positives, where retained customers were incorrectly flagged as churn risks, and 43 false negatives, where churned customers were incorrectly predicted to remain active.

The model achieved an overall accuracy of approximately 89.5%, with strong recall performance for the churned class. From an operational perspective, false negatives represent the costliest outcome because they correspond to customers who churn without receiving retention intervention. The relatively low number of missed churners demonstrates that the baseline model effectively identifies at-risk customers while maintaining balanced performance across both classes. These results establish a reliable benchmark for evaluating more advanced machine learning models in subsequent milestones.

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Classification Report:

              precision    recall  f1-score   support

           0       0.91      0.88      0.89       498
           1       0.88      0.91      0.90       502

    accuracy                           0.90      1000
   macro avg       0.90      0.89      0.89      1000
weighted avg       0.90      0.90      0.89      1000


Accuracy: 0.895
```

Key Insights

The exploratory analysis and baseline modeling produced several important findings regarding customer churn behavior:

- The dataset demonstrates excellent structural quality with no missing values and balanced churn classes, supporting reliable model evaluation.
- Behavioral engagement variables, particularly watch activity and login recency, show stronger predictive influence than demographic attributes.
- Customer inactivity, measured through Last Login Days, consistently emerges as the strongest churn indicator across EDA and model interpretation.
- Extreme engagement outliers represent genuine heavy users rather than data errors and may provide opportunities for customer segmentation.
- Logistic Regression achieved strong baseline performance, while ensemble models demonstrate potential for significantly improved predictive accuracy.

Next Steps

Future work will focus on improving predictive performance and expanding analytical depth through the following steps:

- Evaluate advanced machine learning models such as Decision Tree, Random Forest, and XGBoost to capture complex churn patterns.
- Optimize model performance through hyperparameter tuning, with emphasis on improving recall for churned customers to reduce false negatives.
- Apply dimensionality reduction and feature selection techniques, including PCA, to manage feature expansion after encoding.
- Develop churn probability scoring systems to prioritize customer retention strategies based on risk levels rather than binary predictions.
- Conduct customer segmentation analysis to identify high-risk behavioral groups and support targeted engagement interventions.